

Finite-Sample-Corrected Conformal Test Martingales for Wireless Link Adaptation

Peixuan Wu

George Koudouridis

Genghua Dong

PEIXUAN@KTH.SE

GEORGE.KOUDOURIDIS@ERICSSON.COM

GENGHUA@KTH.SE

Editor: Ernst Ahlberg, Ulf Johansson, Henrik Boström, Alberto Carlevaro, Johan Hallberg Szabadváry and Lars Carlsson

Abstract

Conformal test martingales (CTMs) bet on conformal p-values to accumulate evidence against exchangeability. With finite calibration, a threshold event has finite-grid uncertainty around its nominal probability. We use the Dvoretzky-Kiefer-Wolfowitz inequality to bound this probability and subtract the largest admissible positive excess expected gain. Under the stated continuous score null and a predictable betting schedule, the corrected single-score wealth is a nonnegative supermartingale on the calibration event. A simulated wireless replay illustrates how the corrected evidence process enters an online monitoring pipeline for alarms and retraining decisions.

Keywords: conformal test martingales; online testing; concept drift; delayed feedback; wireless link adaptation

1. Introduction

CTMs turn conformal p-values into sequential evidence against exchangeability (Vovk et al., 2005; Shafer and Vovk, 2008; Vovk et al., 2021; Boström, 2026). Inductive CTMs keep calibration scores fixed during monitoring (Volkhonskiy et al., 2017); for some finite calibration realizations, a threshold bet can then have positive conditional expected gain under the null. We correct this effect in wireless link adaptation, where a predictor selects a modulation and coding scheme (MCS) under changing radio conditions and delayed labels (3GPP, 2020; An et al., 2023; Liu et al., 2023; Balieiro et al., 2021); the corrected evidence process serves both as a change detector and as a trigger for model maintenance.

2. Method

The pipeline starts with fixed calibration samples of scores, one per channel; each online score is compared with its reference sample to obtain a randomized rank conformal p-value, with unusually large scores producing small p-values. A predictable threshold better increases wealth for small p-values and decreases it otherwise, so repeated small p-values accumulate evidence that the stream has left the reference regime.

Score construction. For the immediate feature channel, the feature score is a normalized KNN-distance statistic: with $\mathcal{N}_k(x_t; D_{\text{fit}})$ the k nearest fitting-reference samples to x_t , define $d_{\text{knn}}(x_t; D_{\text{fit}}) = \frac{1}{k} \sum_{x_i \in \mathcal{N}_k(x_t; D_{\text{fit}})} \|x_t - x_i\|_2$. The feature score is normalized by the calibration median and MAD, and the label-dependent score uses the observed label:

$$s_{\text{cov}}(x_t; D_{\text{fit}}) = \frac{d_{\text{knn}}(x_t; D_{\text{fit}}) - \text{med}_{D_{\text{cal}}}}{\text{MAD}_{D_{\text{cal}}}}, \quad \alpha_t^{\text{con}} = -\log p_t(y_t), \quad (1)$$

and is large when the predictor assigns low probability to the observed label. Because the conformal p-value is determined by the score’s rank in its fixed reference sample, a shift that changes the score distribution is directly reflected in the p-value stream; the corrected wealth process converts this score-level incompatibility into sequential evidence. The choice of score therefore determines which distributional changes each channel is sensitive to: the feature score and the label-dependent score provide complementary evidence for covariate changes and predictive-performance changes.

With finite calibration, the threshold event for the empirical p-value has finite-grid uncertainty around its nominal null probability. The Dvoretzky-Kiefer-Wolfowitz (DKW) inequality bounds this event probability by shifting the finite rank grid by $\epsilon = \sqrt{\log(2/\delta)/(2m)}$ (Dvoretzky et al., 1956; Massart, 1990); for every value in this interval, we subtract the largest admissible excess expected multiplier above one. The corrected factor is rescaled to keep the multiplier nonnegative, so the conditional expected multiplier is at most one on the calibration event. Under this fixed-score null and predictable betting, the corrected single-score wealth process is a nonnegative supermartingale on the DKW event.

Radio measurements feed the immediate feature channel and delayed labels feed the label-dependent channel (Vovk, 2020). The corrected evidence process supports alarms and retraining; the formal statement concerns the corrected fixed-score evidence process, and the pipeline shows its use in delayed-feedback monitoring.

3. Experiments

We replayed a simulated 5G MCS stream and compared the corrected CTM with a classical two-stream detector (KSWIN on feature scores (Raab et al., 2020) plus DDM on delayed prediction errors (Gama et al., 2004)). The table records annotated changes found and stable-window alarms, where the latter indicates alarms inside annotated no-change windows.

Table 1: Detector results on the simulated stream.

Detector	Annotated changes found	Stable-window alarms
Uncorrected CTM	3 of 3	Yes
Corrected CTM	2 of 3	No
Classical two-stream	3 of 3	Yes

In this replay, the DKW-corrected CTM is a more conservative alarm signal: it detects two of the three annotated changes while avoiding alarms in stable windows, illustrating the intended effect of reducing spurious evidence accumulation before costly retraining.

4. Conclusion

The main contribution is the finite-sample correction itself, which subtracts the finite-calibration excess allowed by the reference-score envelope; the replay illustrates how the corrected evidence process supports alarms and retraining-oriented operation.

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